

BLDC Motor Report

Group 9

Griffin Laprise, Alexander Moshtagh, Annabelle Kobzar, Jacob Isho

December 9th, 2025

INTRODUCTION

The objective of this project was to design, construct, and test a small brushless DC (BLDC) motor that satisfies the lab constraints of less than 75 W input power, 15 V supply voltage, and 5 A current draw, while adhering to the size constraint of 74mm x 74mm x 44mm. Furthermore, the produced motor should be able to accelerate and maintain a constant speed and torque for ideal operation with minimal vibration. Our design is a radial flux outrunner with a 6-tooth stator and 8-pole rotor, driven from a three-phase delta connection at 12 V and a rated input current of 3 A. The produced motor also had dimensions of 57mm x 57mm x 41mm. During testing the motor had achieved a maximum stable speed of 2360rpm while drawing 0.8 A of input current.

This report encompasses the full design thought process, calculations for all components, CAD design for both the stator and rotor, and assembly of the brushless DC motor as well as the results from testing and overall reflections.

DESIGN PROCEDURE

For this open-ended project each group was given the option to design and construct an axial flux motor or a radial flux, inrunner or outrunner motor. Each group was also tasked with selecting the number of teeth and poles to optimise their motor, as well as which wire thickness would work best and the overall dimensions of their motor.

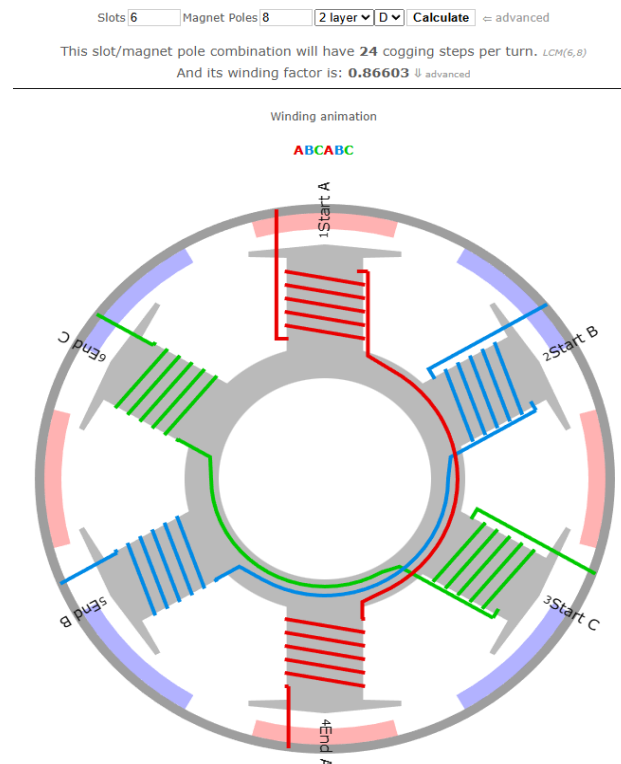
Our group selected to create a radial flux outrunner motor as it aligned closest with the provided design materials and tutorials while maintaining simplicity compared to the other options. Radial flux outrunners are most suited for our use cases when considering the ease of assembly, thermal management, and torque and speed characteristics. As for the ease of assembly, outrunner motors allow for simple coil winding as all coils are diverging from the center, allowing for more space, whereas for inrunner motors the coils are more difficult to manage as they converge towards the center, where it is much tighter. In addition, by making an outrunner motor it allows for much more simple wire management, as all the wires can be spliced and connected at the base of the stator where they all meet. Thorough research concluded that thermal management would be greater for radial flux motors as they are able to have openings at the top and base of the motor, whereas for axial flux motors it is not possible to have openings as that is where the magnets are located.

It was concluded an outrunner motor would be more ideal for our torque and speed applications; this is because outrunner motors work better for higher torque and lower speed applications compared to inrunner motors which are more ideal for higher speed and lower torque applications. A lower speed is ideal because as our rotor is 3D printed, the

plastic used is subject to deformation and possibly even breaking due to the outwards force produced by the magnets while rotating. From these conclusions we had decided a radial flux outrunner motor would best fit our needs while maintaining simplicity.

Once the motor type was decided we moved toward picking the quantity of teeth and poles; from the tutorials we were told to pick a ratio that would provide a cogging ratio near 1, to reduce the ripple torque of the motor. Furthermore, we were also told the quantity of teeth must be a multiple of 3, as there were 3 phases being applied to the motor, and that the quantity of poles had to be a multiple of 2 as this allows for no repeated north and south pole magnets when every other magnet is flipped. It was also vital to ensure the ratio of poles was not a multiple of the quantity of teeth, while being as close as possible. As a ratio that is equal to a multiple of teeth would lead to locking up. A final criterion we were told was that a higher cogging steps per turn number would lead to a smoother motor and was ideal to maximize. From this we concluded that a ratio of 8 poles and 6 teeth would be ideal for our scenario as this provided a cogging ratio near 1 while meeting all other requirements. This allowed for even spacing around the motor as well as maximizing the quantity of poles which would lead to a smoother motor with an easier start up.

Design Software:



In the above figure we used a software to aid us in calculating the cogging ratio, cogging steps per turn, and an optimal wiring scheme for our criteria. From our testing we found the most optimal ratio of teeth to poles resulted in a cogging ratio of 0.86603 which is near 1 and ideal for a motor as it correlates to the efficiency and leads to less ripple current and therefore less jerk within the motor as there would be less torque opposing the rotation of the motor. Furthermore, the cogging steps per turn was calculated to be 24, which correlates to the quantity of magnet alignments per full revolution. A higher cogging steps per turn leads to a smoother motor, this is especially noticeable at lower speeds where each step in the motor is more visible.

Design of Rotor and Stator:

The design of our rotor was with 8 poles in mind as mentioned in the previous steps. This number was also high enough to ensure that there would be no issues with the motor not having a strong enough magnetic field to run which was found to be an issue with some 4 or fewer pole designs where the gaps between the magnets proved to be too large to allow the motor to electromagnetically pull itself to the next magnet in the rotation. In terms of design, it was much easier to start with the rotor as in our case we had less teeth than poles for optimal operation. This meant by designing the rotor first, it was possible to finetune the distance between adjacent magnets while adhering to the overall dimension requirements as the rotor was a larger component than the stator.

The design of the rotor was optimised for strength, weight and airflow. This led to designing the rotor to have a thicker amount of material behind the magnets which prevented not only the magnets from forcing their way through the rotor at high speeds but also preventing warping due to plastic deformation at high speeds. In addition, the receiving output shaft where the output rod was connected to the rotor, was reinforced to prevent the rotor from wobbling in place and improve rotational stability, this was done by thickening the plastic at the intersection to the output shaft to increase the contact patch. All other lesser important portions of the rotor were thinned to reduce weight, and an opening was created on the top of the rotor to allow for airflow to cool and reduce the temperature of the coils on the stator.

As for tolerances, the rotor was designed to be a tight fit on the output shaft to reduce end shake and rotational wobbling. Considering this, the rotor output shaft mating connection was designed to be the same diameter as the rod itself for a press fit. Furthermore, the 8 N52-grade permanent magnets were designed to be 0.2mm larger in all directions to ensure they would fit without complications and allowed for them to be glued

in place later to ensure they are fixed securely. The N52 magnets were chosen for not only their flat thin geometry, which aligns nicely with our rotor design, but also by choosing the N52 magnets which have a stronger magnetic field than the N32 magnets. This allows for a larger air gap while maintaining the same magnetic field strength as the N32 magnet. The magnets were installed with alternating north and south poles between adjacent magnets to create the required alternating magnetic field direction and ensure proper BLDC motor operation.

For the winding configuration it was concluded that the ABC winding scheme in combination with six teeth would be very easy to implement in comparison to LRK or dLRK winding schemes. This meant that the size of the coils on the teeth should be approximately the same size as the magnets themselves to maximize the electromagnetic force produced by the coil while ensuring it is not larger than the magnet itself which would lead to leakage flux.

The design of our stator followed the design of our rotor in that we concluded an air gap of 1.5mm would be ideal as it is well within the 2mm maximum and would prevent any discrepancies in the rotor from rubbing on the stator and lead to friction losses. It was also ideal that the air gap was less than 2mm as it allowed for a stronger magnetic coupling and therefore a more efficient motor, knowing this, the gap of 1.5mm seemed to be a good middle ground. From this the outer diameter of the stator was now calculated at 47mm. The stator itself was designed so that the height and width of the teeth approximated the size of the magnets to produce an efficient motor and align with the ABC winding scheme. Proper tolerances were account for the inside of the teeth to ensure the coils would fit nicely and fill the gap perfectly to ensure a 1:1 ratio for the magnet size to the coil size.

Furthermore, the inlay of the bearings was designed to be the exact size of the bearings themselves to ensure a press fit. A larger inside diameter was also accounted for in the center of the stator to ensure the output shaft has no friction within the stator itself. Once done a final spacer was added to the base to rise the motor above the base plate to allow for easier wire management and to prevent friction of the rotor on the base itself. The spacer was designed as a circle to prevent kinks when bending wires around it and to allow for a simple place to glue and fix the stator to the base. Finally, the stator was printed in two separate halves (see Figures 4 and 5), allowing for a print with no supports which left the surface finish to be smooth to prevent friction losses and for a more precise print. The two halves were later permanently connected with glue.

The concluded CAD designs can be seen in Figures 2 through 6.

CAD:

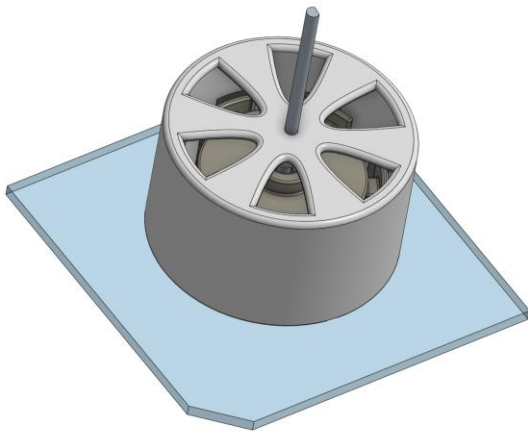


Figure 2: Full assembly of motor

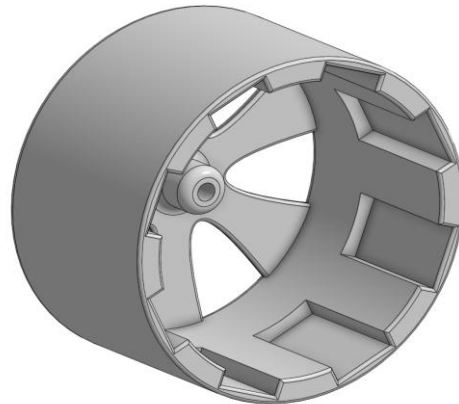


Figure 3: Rotor

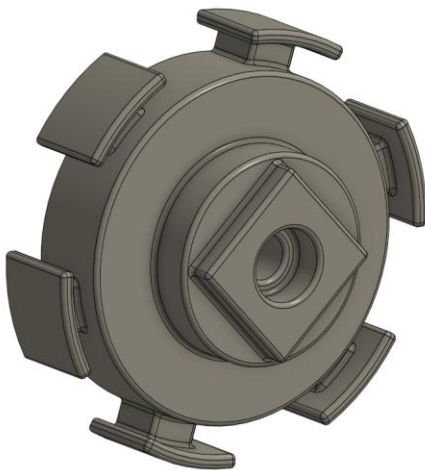


Figure 4: Stator bottom

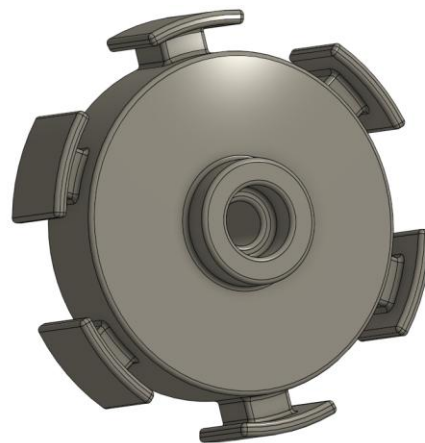


Figure 5: Stator top

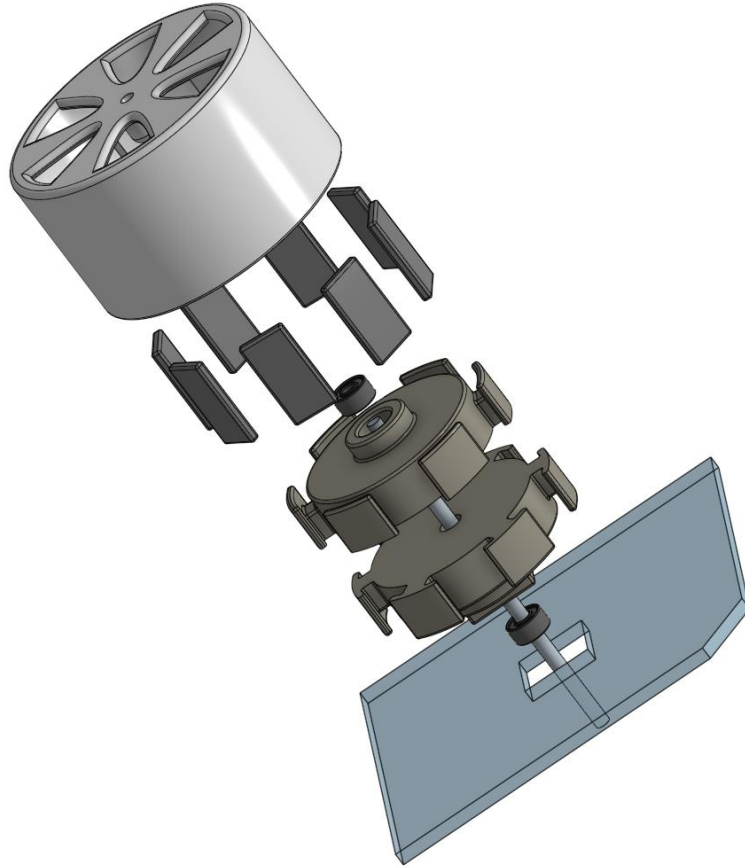


Figure 6: Blown up motor assembly

Calculations:

With the given power constraint of 75 W maximum while maintaining a current draw of 5 A or less and a required voltage of 15 V or less, our motor was designed for a current draw of 3 A and a required voltage of 12 V. This moderate combination allowed for our motor to have a power draw of 36 W, which is much less than the 75 W maximum. Based on these values, we were able to calculate the number of turns of wire per tooth for our specific application.

Given an ideal motor has a 100% efficiency rating, which means it perfectly converts electrical input power to mechanical output power.

$$P_e = P_m$$

This means that the electrical input power $P_e = VI$ and the output mechanical power $P_m = \tau\omega$, are of the same value.

It was decided our motor would have a torque τ value of 0.2Nm which correlated nicely to the size of our motor.

$$\tau = 0.2Nm$$

From this the speed was deduced to be $\omega = \frac{180rad}{s}$ from subbing the power of 36W and the torque of 0.2Nm into the mechanical power equation. Which correlates for a revolutions per minute of approximately 1700rpm .

To maintain power drawn, when the speed reduces, the torque increases, and vice versa. Due to the smaller nature of our BLDC motor, we decided to use a higher speed on correlation to our torque produced. This is ideal because a smaller motor has less weight to move and therefore requires less torque to do so, therefore a higher speed is produced.

Choosing the N-52 permanent magnets, correlates our residual flux density value to $B_r = 1.42T$.

The measurement from the CAD provides the following values:

- Magnets thickness: $t = 2mm$
- Tooth wall thickness: $k = 1.5mm$
- Air gap thickness: $g = 1.5mm$
- Length of stator: $L = 0.0202m$
- Radius of stator: $R = 0.0235m$

From these values we can calculate the flux density at the surface of the stator.

$$B_a = B_r \frac{t}{t + g + k} = 1.42T \frac{0.002}{0.002 + 0.002 + 0.0015} = 0.5164T$$

The flux density at the surface of the stator (B_a) can be used in tandem with the length of the stator (L), radius of the stator (R), input current (i), and the quantity of teeth per phase (n), to calculate the number of turns per tooth (N).

$$N = \frac{\tau}{4 \cdot n \cdot B_a \cdot L \cdot R \cdot i} = \frac{0.2}{4 \cdot 2 \cdot 0.5164 \cdot 0.0202 \cdot 0.0235 \cdot 3} = 33.99 \approx 34$$

This value correlates to the required quantity of turns per tooth at N=34, which is well with the minimum of 5 turns and maximum of 50 turns. Therefore, to meet our design requirements, we would require exactly 34 turns per tooth.

Considering the expected input current of 3 A, the wire thickness of 0.42mm was used over the 0.224mm as the larger wire has a greater current carrying capacity than the smaller wire. Checking the current carrying capacity of the larger wire results in a value of 3.3 A, which is larger than our expected 3 A of current draw and therefore is a great fit.

This value was then cross checked against the CAD design to ensure that quantity would fit properly on the teeth of the stator by considering how many turns of wire can fit side to side and how many turns high.

RESULTS:

The overall testing of our motor was a success, and we were able to achieve powered rotation on the motor. The assembled motor can be seen in Figure 7, 8 and 9. We ran into no problems during the testing which was ideal. Once testing began, the motor accelerated quickly and we were able to achieve a maximum stable speed of 2360rpm at a current draw of 0.8 A. Seeing that our motor ran at this speed consistently, and without issue, it shows the for though we put into design process, including both the calculations and CAD. In addition, our motor proved to be efficient considering how smoothly it ran and showed minimal room for improvement due to the lack of audible vibrations or overheating during the testing process. Overall, the motor performed as expected during the testing. When the testing was complete, all portions of the motor still fell within the tolerances we had set, and all press fit components were still tight and there was no visible warping.

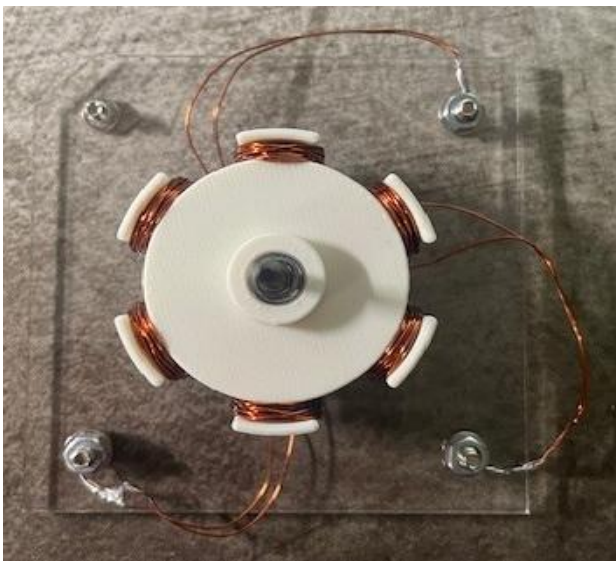


Figure 7: Stator and coils

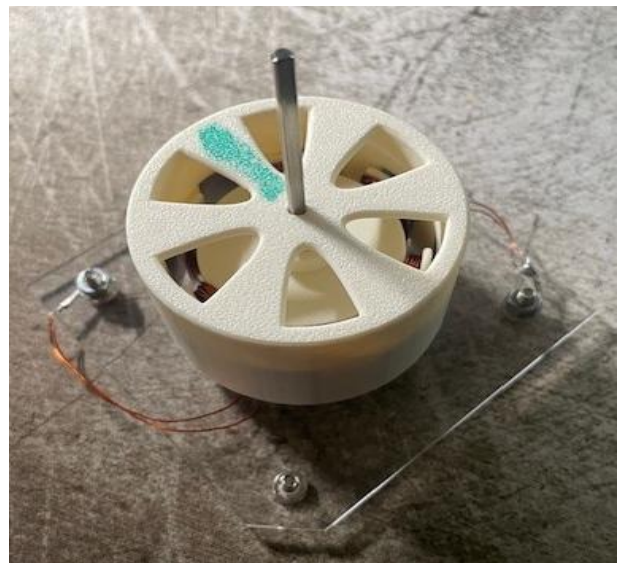


Figure 8: Fully assembled motor

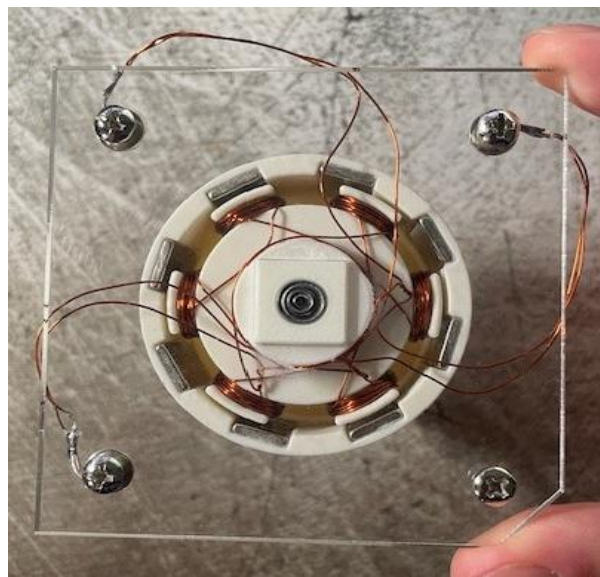


Figure 7: Base of the motor and wire management of the three phases

DISCUSSION:

The outcomes of the built-in BLDC motor closely match the calculations and design specifications given throughout the report. The chosen 6-tooth, 8-pole outrunner topology effectively generated smooth rotational motion, confirming the design considerations regarding air-gap size, magnetic alignment, and cogging ratio. The motor operated significantly below the intended 3 A input current, as indicated by the measured maximum stable speed of 2360 rpm at only 0.8 A. This suggests that the design was conservative and that the electromagnetic geometry supplied more than enough torque for rotation at low load. Additionally, the absence of audible vibration or hotspot formation during testing shows that the magnet spacing, coil placement, and rotor–stator tolerances were correctly applied.

The earlier flux density and torque calculations, which predicted roughly 1700 rpm under nominal loading, are also supported by the motor's performance. It is possible that the actual mechanical load was lower than the theoretical design load, resulting in a higher-than-expected rotational speed, if speeds exceeding this were achieved while drawing noticeably less current. Furthermore, the structural design decisions such as incorporating airflow openings and adding reinforcement behind the magnets, proved successful because, following testing, there was no deformation, rubbing, or detachment. Real-world losses like bearing friction, coil winding tolerances, and variations in the real magnet strength relative to idealized models can all be blamed for small differences between the measured and predicted speeds.

Overall, the motor's behavior was in line with both the design goals and electromagnetic theory. The selected dimensions, winding count, and magnetic configuration were well-optimized for the project's constraints, as evidenced by the strong correlation between analytical predictions, CAD implementation, and physical test results.